

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations**

May - June 2023

20ME601 – INDUSTRIAL ROBOTICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) What are the different robot configurations classified by different industrial association?	10	L*2	1	1
	b) A cylindrical robot has a prismatic joint with a range of travel of 800mm. The control memory for this joint has a 10-bit capacity. It has been recorded that the associated mechanical inaccuracies with the said arm show a random distribution of random variable of the robot position about the mean position of the taught point gives a standard deviation of 0.1mm the standard deviation is equal in all direction. Determine the following, i. The control resolution for the axis. ii. The spatial resolution for the prismatic joint. iii. The accuracy defined. iv. The repeatability of the robot link.	10	L3	1	1
2.	a) What are different robot configurations and describe the types of robot joints?	10	L1	1	1
	b) A double acting single ended piston hydraulic cylinder is used to actuate one of the linier arm joints of a Cartesian robot. The diameter of a piston is 10cm & the diameter of the rod is 5cm. A pump supplies hydraulic oil at a rate of 36cm ³ /sec with a pressure of 40N/cm ² . determine i. The force that can be applied by the piston in forward and reverse strokes ii. The maximum velocity with which rod can operate in forward and reverse directions	5	L3	1	1
	c) A DC motor used to drive a robot joint has a torque constant, 1.25 N-m/A and the voltage constant of 12×10^{-3} V/rpm. The armature resistance of 2.5 ohms. A voltage of 25V is applied at a point of time of robot cycle when the joint is stationary. Determine, i. The torque of the motor immediately after the voltage is applied. ii. The back e.m.f and the respective torque corresponding to rotational speeds of 250 rpm, and 500 rpm.	5	L3	1	1
		Unit – II			
3.	a) Explain the concept of direct and inverse kinematics with an example?	5	L2	2	1
	b) Explain the concept of pre & post multiplication with an example.	5	L2	2	1
	c) Illustrate with usual notations the derivation of rotation of body attached frame with respect to arbitrary axis by an angle θ .	10	L3	2	1
4.	a) Define and illustrate the link and joint parameters.	10	L2	2	1

P.T.O.

- b) A six joint robotic manipulator equipped with a digital TV camera is capable of continuously monitoring the position and orientation of an object. The position and orientation of the object with respect to camera is expressed by a matrix T_1 , the origin of the robot's base co-ordinate with respect to the camera is given by T_2 , and the position and orientation of the gripper with respect to the base co-ordinate frame is given by T_3 .

$$T_1 = \begin{bmatrix} 0 & 1 & 0 & 5 \\ 1 & 0 & 0 & 6 \\ 0 & 0 & -1 & 10 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_2 = \begin{bmatrix} 1 & 0 & 0 & -20 \\ 0 & -1 & 0 & 10 \\ 0 & 0 & -1 & 12 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad T_3 = \begin{bmatrix} 1 & 0 & 0 & 8 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Determine,

- I. The position and orientation of the object with respect to the base co-ordinate.
- II. The position and orientation of the object with respect to gripper.

10 L3 2 1

Unit – III

5. a) With the block diagram, transfer function & features explain Proportional-Integral (PI), Proportional-Derivative & Proportional-Integral-Derivative Controller.
- b) The second joint of a SCARA manipulator is required to move from $\theta_2=30^\circ$ to 150° in 5 seconds. Find the cubic polynomial to generate the smooth trajectory for the joint. What is the maximum velocity and acceleration for this trajectory?

10 L1 3 1

10 L2 3 2

6. a) Obtain the transfer function of a DC motor driven robot joint.
- b) Explain briefly the steps in trajectory planning.

10 L3 3 1

10 L2 3 1

Unit – IV

7. a) Explain the functions of the machine vision system using a block diagram.
- b) With a neat sketch explain the construction and working principle of the vidicon camera.

10 L1 4 1

10 L2 4 1

8. a) Briefly explain analog to digital conversion process.
- b) Explain briefly the principle functions of a robot vision system.

10 L2 4 1

10 L1 4 1

Unit – V

9. a) Illustrate several robot programming approaches.
- b) Write a VAL-II program to pick the parts from pick-up point and palletize them in a pallet of 5 rows and 5 columns. The size of each pallet is 50 units x 50 units.
10. a) Explain the concept of interlock with an example.
- b) Explain the different VAL 2 motion control, speed control, and position control commands.

10 L1 5 1

10 L4 5 2

10 L1 5 1

10 L2 5 1